

Research on Text Recognition Method of Financial Documents

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ABSTRACT

Aiming at the problems of text box overlap and missing text content in the recognition of financial documents in the Detecting Text in Natural Image with Connectionist Text Proposal Network(CTPN) model, it is proposed to improve the CTPN network structure based on the Inception network and optimize the feature extraction process. At the same time, the confidence loss function is introduced in the classification to optimize the candidates. Selection of the position of the frame anchor. A comparative experiment with the CTPN model proves that the improved network model has a certain improvement effect on text target detection. Created and implemented a text recognition model of DenseNet+Bi-directional Long Short-Term Memory(BLSTM)+CTC. The DenseNet network establishes a connection between feature maps, achieves the reuse of features, reduces the amount of parameters, and reduces the amount of calculations for each layer of the network. BLSTM can associate contextual information to assist prediction. The CTC loss function solves the text prediction and recognition when the input and output are not aligned. Experiments show that the model designed in this paper has a certain improvement in the recognition performance of traditional recognition systems and CRNN models.

CCS CONCEPTS

• Computing methodologies; • Artificial intelligence; • Computer vision; • Image and video acquisition;

KEYWORDS

Optical character recognition, deep learning, convolutional neural network, text recognition

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1 INTRODUCTION

In recent years, image optical symbol recognition (OCR) has attracted more and more attention in the field of computer vision due to its practical applications in many fields. OCR text recognition

technology is composed of two core subtasks: text detection and text recognition. For the text content of the location, the commonly used model algorithms mainly include: text detection: DB [2], EAST [3], CTPN [4], RRD [5], etc.; text recognition: CRNN [6], StarNet [7], Rosetta [8], etc.

With the continuous development of deep learning in recent years, OCR technology based on convolutional neural networks has broken the framework of traditional OCR technology [1], DB [2], EAST [3], CTPN [4], RRD [5], etc. based on depth The learned text detection algorithm proposes a qualitative leap in recognition rate and accuracy. CTPN [4] combined with CNN and LSTM deep network can effectively detect the horizontally distributed text in complex scenes, but the detection of this method is very poor in vertical text or oblique text. EAST uses FCN network + Locality-Aware NMS to effectively solve the problem of vertical and oblique text, but the effect is not good in long text recognition. With the continuous development of deep learning in recent years, OCR technology based on convolutional neural network breaks the framework of traditional OCR technology [1]. DB [2], East [3], ctpn [4], RRD [5] and other text detection algorithms based on deep learning are proposed to have a qualitative improvement in recognition rate and accuracy. CTPN [4] combined with CNN and LSTM depth network can effectively detect the horizontally distributed text of complex scenes, but the detection of vertical text or inclined text is very poor. East [3] using FCN Network + Locality-Aware NMS can effectively solve the problems of vertical and inclined text, but the effect is not good in long text recognition. In terms of text detection, this paper uses the CRNN algorithm framework for text recognition detection. CRNN [6] is characterized by the fusion of sequence features and convolution features. In terms of text recognition, deep residual network (ResNet) [9] solves the problems faced by deep networks to a certain extent by building connections between different network layers and deepening the network depth. The DenseNet [10] network is based on this, by establishing a connection between a certain layer in the network and all the previous layers, it makes full use of the characteristic information, and improves the quality of the model while using fewer parameters and reducing the amount of calculation. BLSTM [11] is proposed on the basis of LSTM [12], which consists of two LSTM networks in different directions. For a text content, you can use forward LSTM from front to back to solve the dependency of the previous content on the following content, and then use the backward LSTM from back to forward to solve the dependency of the latter content on the previous content, and then two Two networks are connected to the same output layer. CTC [13] is a method based on Loss calculation. The training samples required by CTC instead of traditional Softmaxloss can be trained without alignment. CTC loss can solve the problem of inconsistent prediction results and labels, and performs better in document recognition.

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